Machine Learning HW2

ML TAs
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Machine learning HW2 . Speech

Outline

Task introduction,

- Task Introduction
- Dataset & Data Format Data .
- Submission & Grading .

Task Introduction

Task Introduction

1. Data Preprocessing: Extract MFCC features from raw waveform (already done by TAs!)
2. Classification: Perform framewise phoneme classification using pre-extracted MFCC features

waveform

data

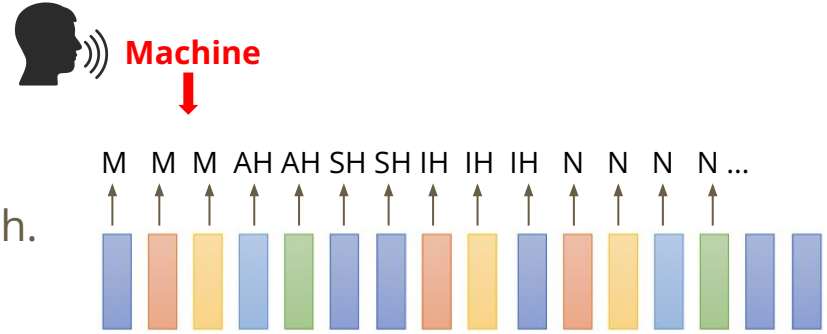
frame

Task Introduction

Task: Multiclass Classification

Framewise phoneme prediction from speech.

Multiclass classification



train. data . dim d.

What is a phoneme?

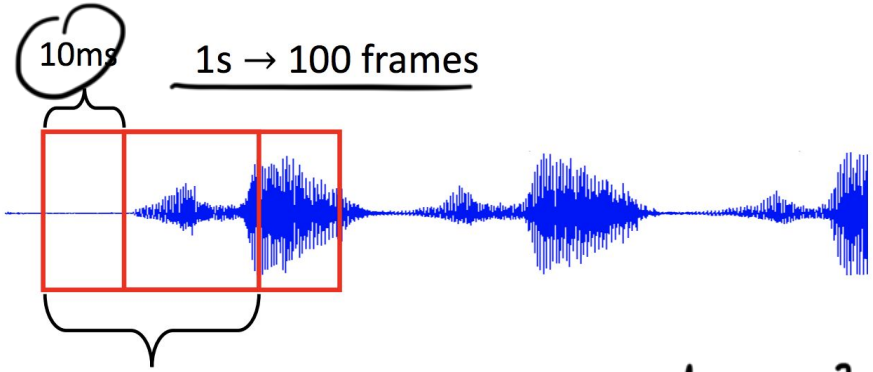
A unit of speech sound in a language that can serve to distinguish one word from the other.

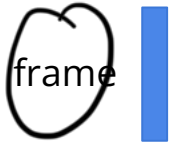
MAHSHIHN'

- bat / pat , bad / bed
- Machine Learning → M AH SH IH N L ER N IH NG

M AHSH LERN.

Data Preprocessing





39-dim MFCC
80-dim filter bank
...

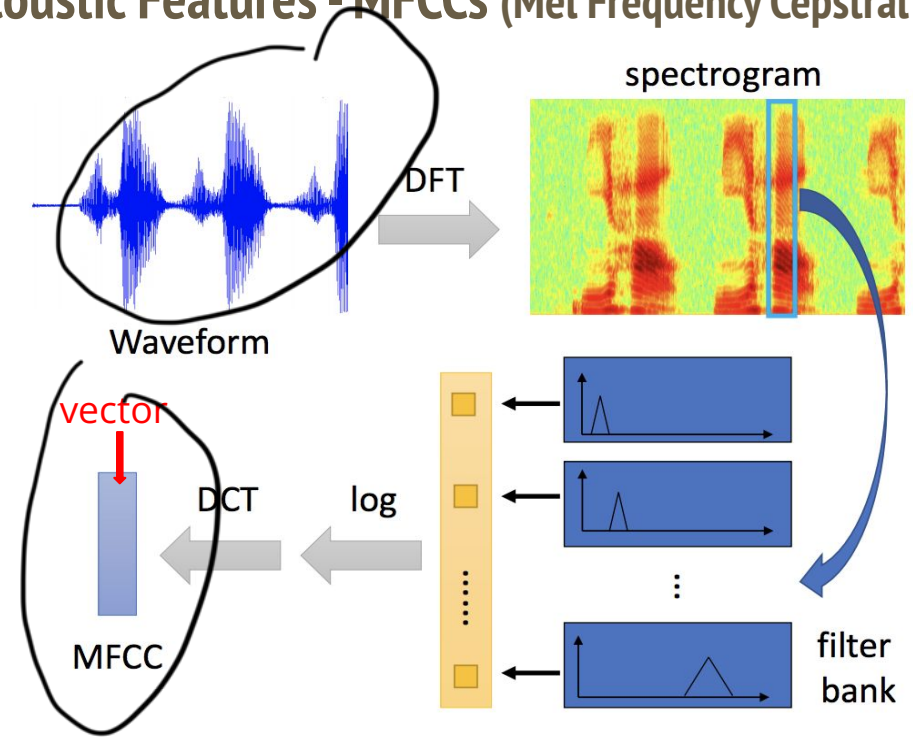
dim 39.

a vector

length T, dimension d

Data Preprocessing

Acoustic Features - MFCCs (Mel Frequency Cepstral Coefficients)



For more details,
please refer to Prof. Lin-Shan Lee's
[Introduction to Digital Speech Processing] Chap.7

Image ref.
Prof. Hung-Yi Lee
[2020Spring DLHLP] Speech Recognition

More Information About the Data

25

Since each frame only contains 25 ms of speech, a single frame is unlikely to represent a complete phoneme

phoneme.

- Usually, a phoneme will span several frames
- Concatenate the neighboring phonemes for training

Finding testing labels or doing human labeling are strictly prohibited!

39

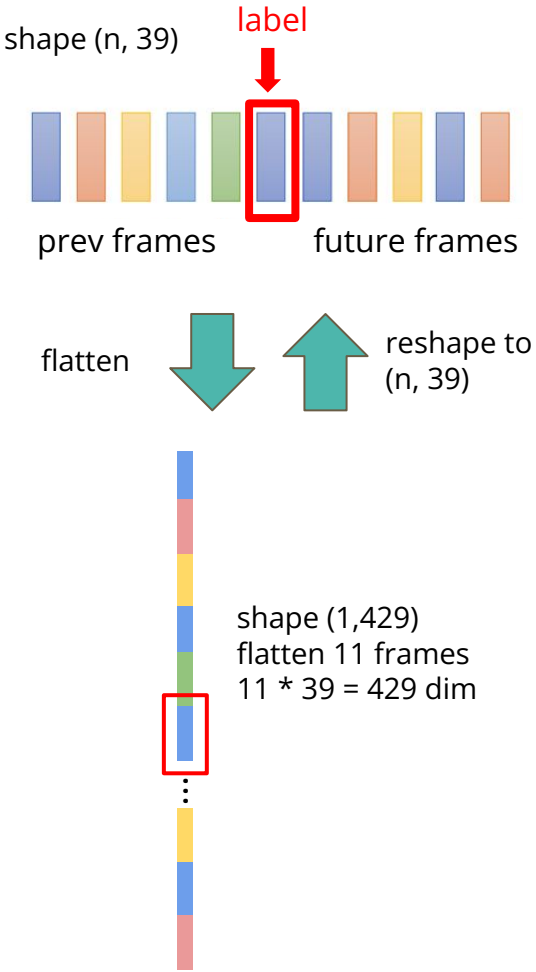
(1x39)

dim = 1,

[ ] [ ]

Concatenate

Concatenate.



Introduction to Digital Speech Processing

dim = 0 flatten!

Dataset & Data Format

Data

$f_1 \quad f_2 \quad \cdots \quad f_{38} \quad y_1 \quad \cdots \quad y_{41}$

Dataset

Dataset LibriSpeech.

LibriSpeech (subset of train-clean-100)

- Training: 4268 preprocessed audio features with labels (total 2644158 frames)
- Testing: 1078 preprocessed audio features (total 646268 frames)
- Label: 41 classes, each class represents a phoneme

2644158 個 train data? → train.

→ $y_1 \ldots y_{p_0}$

1078 test data

y 41 class.

41 phoneme.

Data Format

注意：这两个数据一样了

- Data Format (The TAs have already extracted the features)

libriphone/

- train_split.txt (train metadata)
- train_labels.txt (train labels)
- test_split.txt (test metadata)
- feat/
  - train/
  - test/

test metadata 待定.

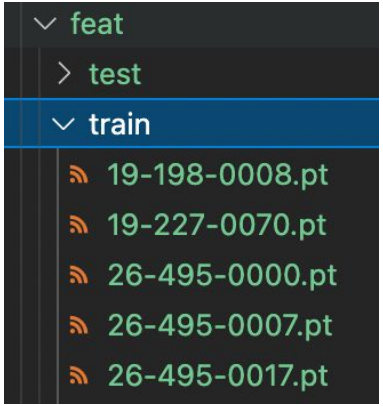
train_split.txt

```
2007-149877-0023
60-121082-0044
5688-41232-0018
1898-145702-0018
307-127539-0028
6147-34607-0009
7226-86965-0049
6078-54007-0024
5688-15787-0042
5808-48608-0055
```

train_labels.txt

```
2007-149877-0023 0 0 0 0 0 0 0 0
60-121082-0044 0 0 0 0 0 0 0 0 0
5688-41232-0018 0 0 0 0 0 0 0
1898-145702-0018 0 0 0 0 0 0 0 0
307-127539-0028 0 0 0 0 0 0 0
6147-34607-0009 0 0 0 0 0 0 0
7226-86965-0049 0 0 0 0 0 0 0
6078-54007-0024 0 0 0 0 0 0 0
5688-15787-0042 0 0 0 0 0 0 0
5808-48608-0055 0 0 0 0 0 0 0
```

features: 39-dim MFCC w/ CMVN
{filename}.pt for each utterance(audio)



feat.

Using additional data is prohibited. Your final grade will be multiplied by 0.9!

Data Format

T, 39

- Each .pt file is extracted from one original wav file
- Use torch.load() to read in .pt files as torch tensors
- Each tensor has a shape of (T, 39)

T, 39

T f

39 dims

Causal concatenate

T frames

```
tensor([[-0.9555, -0.9062,  0.9451,  ..., -1.4516, -1.5912, -1.3270],
        [-0.9434, -0.9633,  0.7211,  ...,  0.1566, -0.0150, -0.1353],
        [-0.8907, -0.9749,  0.6556,  ...,  1.1867,  0.4603, -0.0459],
        ...,
        [-1.0778, -0.7979,  0.8335,  ...,  0.6452, -0.3527, -0.7415],
        [-1.1911, -1.0670,  0.6462,  ...,  0.3025, -0.6755, -0.9707],
        [-1.1044, -1.0259,  0.7016,  ..., -0.1956, -0.4646, -0.5964]])
```

Submission & Grading

Submission & Grading

- Leaderboard (4%): Kaggle
- Code submission (2%): NTU COOL
- Report submission (4%): Gradescope

Kaggle Public Baselines

- (1%) Simple baseline: 0.45797 (sample code)
- (1%) Medium baseline: 0.69747 (concat n frames, add layers)
- (1%) Strong baseline: 0.75028 (concat n, batchnorm, dropout, add layers)
- (1%) Boss baseline: 0.82324 (sequence-labeling(using RNN))

For boss baseline, you can refer to pervious course recording of RNN: video 1 and video 2.

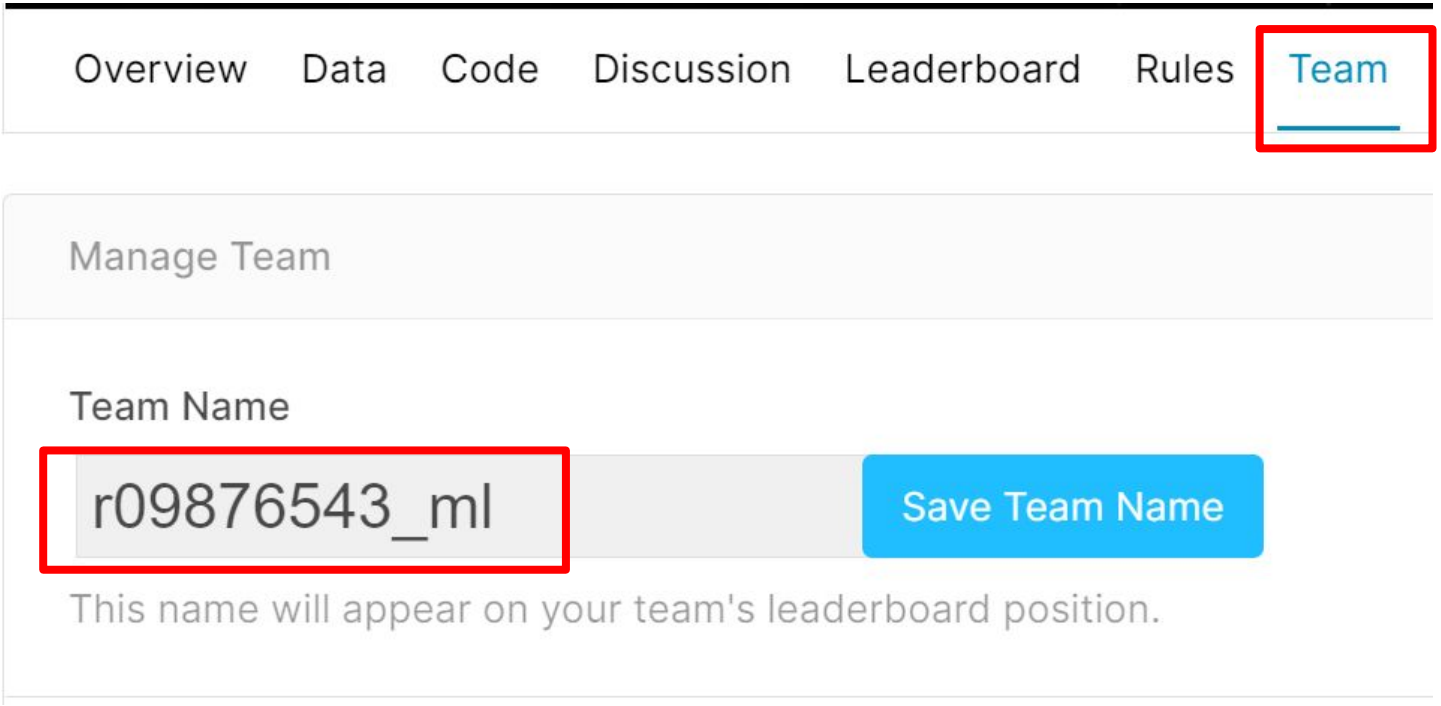
Kaggle Submission

Kaggle Link: https://www.kaggle.com/c/ml2022spring-hw2

- Displayed name: **`<student ID>_<anything>`**
  - e.g. `b06901020_puipui`
- You do NOT have to change your account name, just modify “team name” under the “team” tab
- Submission format: **`.csv`** file
- Evaluation metric: accuracy
- Submission deadline:
  - **2022/3/18 23:59 (UTC+8)**

```
Id,Class
0,0
1,0
2,0
3,0
4,0
```

Kaggle Submission



Kaggle Submission

- You may submit up to **5** results each day (UTC).
- Up to **2** submissions will be considered for the private leaderboard.

prediction_large.csv 2 years ago by ntuee_jizz model_large3_684_compressed.pth, size = 201KB, params: 93139 (rabbit ensemble)	0.65059	0.66341	☑
prediction_large.csv 2 years ago by ntuee_jizz model_large3_676_compressed.pth, size = 201KB, params: 93139 (rabbit ensemble)	0.65282	0.65422	☐
prediction_large.csv 2 years ago by ntuee_jizz model_large2_669_compressed.pth, size = 222KB, params: 103623	0.65394	0.65254	☑

remember to select **2** results for your final scores before the competition ends!

Code Submission (2%)

- Compress your code and report, then submit it to NTU COOL.

`<student ID>_hw2.zip`

`e.g. b06901999_hw2.zip`

- We can only see your last submission.
- Do not submit your model or dataset.
- If your code is not reasonable, your final grade will be multiplied by 0.9!
- Submission deadline:
    - **2022/3/18 23:59 (UTC+8)**

Report Questions

1. (2%) Implement 2 models with approximately the same number of parameters, (A) one narrower and deeper (e.g. hidden_layers=6, hidden_dim=1024) and (B) the other wider and shallower (e.g. hidden_layers=2, hidden_dim=1700). Report training/validation accuracies for both models.

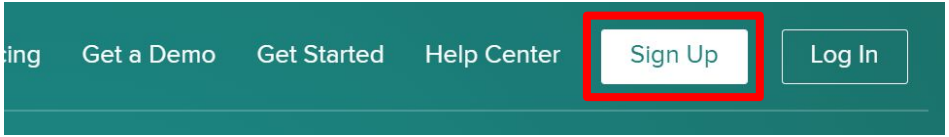
2. (2%) Add dropout layers, and report training/validation accuracies with dropout rates equal to (A) 0.25/(B) 0.5/(C) 0.75 respectively.

Report Submission

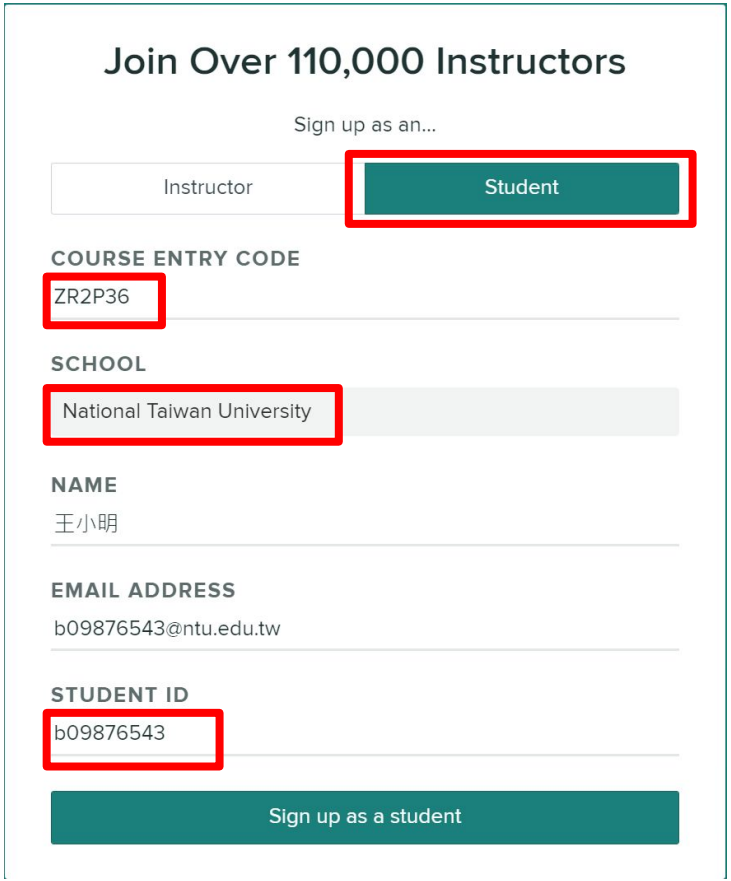
- Submit with gradescope, no need to upload any files.
- We can only see your last submission.
- Submission deadline:
  - **2022/3/18 24:00 (UTC+8)**

Gradescope Registration

1.



2. Entry Code: ZR2P36



3.

Welcome to Gradescope! You are now enrolled as a student in 921 U2

To get started, you will first need to set your password (link will expire (CST)).

Gradescope Submission

NAME	STATUS	RELEASED	DUE (CST)
ML2022Spring HW2	No Submission	FEB 23	3 weeks, 2 days left MAR 19 AT 12:00AM

ML2022Spring HW2

Q1

2 Points

Implement 2 models with approximately the same number of parameters, (A) one narrower and deeper (e.g. hidden_layers=6, hidden_dim=1024) and (B) the other wider and shallower (e.g. hidden_layers=2, hidden_dim=1700). Briefly discuss the difference in training processes and training/validation accuracies.

Regulations

(*) Academic Ethics Guidelines for Researchers by the Ministry of Science and Technology

- You should NOT plagiarize, if you use any other resource, you should cite it in the reference. (*)
- You should NOT modify your prediction files manually.
- Do NOT share codes or prediction files with any living creatures.
- Do NOT use any approaches to submit your results more than 5 times a day.
- Do NOT use additional data or pre-trained models.
- Your **assignment will not be graded** and your **final grade x 0.9** if you violate any of the above rules.
- Prof. Lee & TAs preserve the rights to change the rules & grades.

If you have any questions, you can ask us via...

- NTU COOL (recommended)
  - https://cool.ntu.edu.tw/courses/11666
- Email
  - mlta-2022-spring@googlegroups.com
  - The title should begin with “[hwX]” (X is the homework number)
- TA hour
  - To be determined